**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 242 · Thinking logically: decisions and program flow

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why must a booking be saved before confirmation?

Q2. What is true AND false?

Q3. What does an if condition control?

Q4. What is false OR true?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

condition — Expression evaluated as true or false.

branch — Path selected by a decision.

AND — True only if both operands are true.

OR — True if either or both operands are true.

truth table — All Boolean input combinations and outputs.

## New learning

Locate the decision — A discount needs BOTH an eligible membership and a basket of at least £50.

Follow the true branch — For member=true and basket=60, the condition is true and 10% discount produces £54.

Follow the false branch — For member=false with basket=60, the AND expression is false, so the full £60 is charged.

## Worked example

A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

1. Predict first — The age condition is true, but the ticket condition is false.

2. Model the reasoning — AND requires both true; the result is false, so the deny branch executes.

# We Do · Practise with guidance

A challenge unlocks when score≥80 OR teacherOverride=true. Trace score=72 with override.

```
bool permitted = score>=80 || teacherOverride;
```

Predict first

The score condition is false because 72<80.

Model the reasoning

The override is true, so OR evaluates true and the challenge unlocks.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Give A AND B for (false,false), (false,true), (true,false), (true,true).

### Question 2

Access is allowed if Year13 AND (HasCode OR TeacherApproved). Evaluate Year13=true, HasCode=false, TeacherApproved=true; then Year13=false, HasCode=true, TeacherApproved=true.

### Question 3

A developer changes the discount condition `member && basket>=50` to `member || basket>=50`. Give two wrong outcomes and correct the expression.

# Exit Ticket · Answer before checking

Replacing AND with OR in a two-part eligibility rule can permit extra cases.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Confirmation must reflect a successful stored change.

Q2. False.

Q3. Which branch of the program runs.

Q4. True.

### You Do pages · Independent questions

Q1. [4] False [1], false [1], false [1], true [1].

Q2. [6] Case 1: bracket false OR true=true [1], true AND true=true [1], allowed [1]. Case 2: bracket true OR true=true [1], false AND true=false [1], denied [1].

Q3. [6] A member spending under £50 incorrectly gets discount [2]; nonmember spending ≥£50 incorrectly gets discount [2]; correct member && basket>=50m [2].

### Exit Ticket · Model answer

True. OR is true when either condition holds.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**